Emily Louie

emily.louie.ca emilyslouie@gmail.com github.com/emilyslouie linkedin.com/in/emilyslouie

EDUCATION

University of Waterloo – *Systems Design Engineering (Bachelors of Applied Science)* **2019 – 2024**

Courses: Data Structures and Algorithms (C++), Digital Systems (C) Cumulative GPA: 89% (3.95/4.0)

TECHNICAL SKILLS

Frontend: JavaScript/TypeScript, React, Redux, Gatsby, HTML, CSS/SCSS, GraphQL, REST

Backend: Java, Spring Boot, JUnit, C/C++, Golang, Python, Docker, Firebase, Azure, Neo4j, PostgreSQL, MongoDB

3D Graphics and Design: Unity, C#, Three.js, AFrame, Figma, Sketch, Adobe Creative Suite, Blender

EXPERIENCE

Unity Technologies – *Software Engineering Intern* **May – Aug 2021**

- Constructed and deployed 5+ advertising experiments on Unity's ad network to **10+ million users**
- Generated **5% more revenue** by creating custom frontend layouts and animations in **TypeScript**, and **React**
- Tracked metrics among 10+ ongoing A/B testing experiments by implementing numerous experiment policies in **Golang** to plan for upcoming features and experiments on the team's roadmap
- Optimized for best development practices by refactoring 10+ custom components and styles to create a component library that allows for the use of Storybook and Linaria

UW Blueprint – *Software Developer* **Aug 2020 – Present**

- Created a full stack starter application with **MERN** and **Flask** for teams to develop software for nonprofits
- Implemented **GraphQL** and **REST** APIs with CRUD operations for **MongoDB** and **PostgreSQL**
- Designed a decision-making platform (arbitrium.web.app) that helped allocate **$2+ million** to local charities

Ontario Digital Service – *Software Engineering Intern* **Sep – Dec 2020**

- Developed and launched 3 new versions of Ontario's COVID-19 data visualizations (covid-19.ontario.ca/data) and informational microsites using **JavaScript** and **React**, averaging **450,000+ unique views** per month
- Reduced average build times by **52%** by optimizing data collection algorithms in the project's build process
- Led the creation of data schemas for Ontario's public COVID-19 datasets used by local news (data.ontario.ca)
- Architected the frontend of ServiceOntario's future appointment booking system using **React** and **Firebase**, which will reduce wait times by **90%** and be used by **5+ million people** annually

Ontario Digital Service – *Product Design Intern* **Jan – Apr 2020**

- Collaborated with engineering teams to pioneer AODA/WCAG 2.0 compliant designs for Ontario's COVID-19 sites (covid-19.ontario.ca) which had **1.25+ million views** the month after launch
- Redesigned a web platform to help **5+ million users** renew their government identification cards annually

Microsoft – *Software Engineering and Product Design Intern* **Jul – Aug 2019**

- Developed the frontend of an internal inventory tracker used by **2,000+ employees** and reduces the request processing time by over **85%**, using **JavaScript**, **React**, and **Azure**
- Presented the beta launch of the app to **800 executives and employees** at the annual general meeting
- Prototyped a mixed reality office wayfinding application for the Hololens using **Unity** and **C#**

PROJECTS

Six Degrees of Spotify – *Java, Spring Boot, Neo4j, Docker, Vaadin, JUnit* **/six-degrees**

- Built a full stack web application to explore that artists are connected by six or fewer song collaborations by maintaining a Neo4j database with **450,000** collaborations parsed from the Spotify API
- Created unit and integration testing suites with **JUnit** to ensure a quality user experience